

Year 7 · Chapter 6 · Geometrical figures: answers (teacher copy)

6.01 Transforming shapes (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: mirror image on the right of the line
- Ex 1b: moved 5 right and 3 down
- Ex 2a: turned upside down through O
- Ex 2b: turned a quarter turn clockwise
- Ex 3a: mirror image below the line
- Ex 3b: turned three-quarters clockwise

Easy (drill)

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|-------------|--------------|--------------|
| 1. (-2, -9) | 10. (-2, -6) | 19. (-4, -1) |
| 2. (3, -7) | 11. (-2, -1) | 20. (3, 2) |
| 3. (10, -2) | 12. (-6, -2) | 21. (-1, 2) |
| 4. (0, -4) | 13. (8, -6) | 22. (6, 2) |
| 5. (8, -6) | 14. (2, 3) | 23. (6, 6) |
| 6. (-1, -1) | 15. (7, -6) | 24. (7, 4) |
| 7. (2, 7) | 16. (8, -4) | 25. (6, 4) |
| 8. (-1, 5) | 17. (-3, -1) | |
| 9. (-6, 0) | 18. (-5, -8) | |

Medium (drill)

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|-------------|-------------|--------------|
| 1. (4, -1) | 6. (-4, 3) | 11. (-4, -1) |
| 2. (-5, 2) | 7. (5, 5) | 12. (6, 4) |
| 3. (2, -6) | 8. (6, 3) | 13. (-2, -5) |
| 4. (-4, -5) | 9. (-4, 4) | 14. (6, 1) |
| 5. (-6, 6) | 10. (4, -2) | 15. (3, 6) |

Challenging

- rotated 270° clockwise
- reflected in the diagonal

Extension

- a rotation of 180° about the point where the lines cross
- Yes: both end in the same position

Exit ticket

- Level 1: reflection
- Level 2: (6, 2)
- Level 3: it points right

6.02 Combined transformations (booklet p. 9)

WE DO examples (teacher)

- Ex 1a: translated, then flipped over the line
- Ex 1b: rotated, then moved 3 left
- Ex 2a: reflected, then moved down 1
- Ex 2b: Often yes: the images end in different places
- Ex 3a: three images drawn in order
- Ex 3b: e.g. rotate 180° , then translate

Easy (drill)

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|----------------|--------------|--------------|
| 1. reflection | 8. 12 left | 15. 5 right |
| 2. reflection | 9. 10 right | 16. 11 right |
| 3. translation | 10. 4 right | 17. 8 left |
| 4. rotation | 11. 14 right | 18. 7 left |
| 5. rotation | 12. 4 left | 19. 1 right |
| 6. rotation | 13. 2 right | 20. 6 left |
| 7. 2 right | 14. 6 right | 21. 12 left |

Medium (drill)

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|------------|------------|--------------|
| 1. (4, -5) | 5. (4, -1) | 9. (0, 5) |
| 2. (8, -3) | 6. (-2, 5) | 10. (-2, -4) |
| 3. (4, 5) | 7. (7, -5) | 11. (-5, -2) |
| 4. (-2, 4) | 8. (-4, 5) | 12. (-5, 3) |

Challenging

- e.g. rotate 180° then translate
- reflect in a vertical line; or reflect then translate

Extension

- (-1, -5)
- (-4, -1)

Exit ticket

- Level 1: the image after one transformation
- Level 2: (4, 6)
- Level 3: (0, -3)

6.03 Line symmetry (booklet p. 15)

WE DO examples (teacher)

- Ex 1a: 2
- Ex 1b: 4
- Ex 2a: 6
- Ex 2b: 0
- Ex 3a: 2
- Ex 3b: infinitely many

Easy (drill)

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|-------|-------|-------|
| 1. 1 | 11. 1 | 21. 1 |
| 2. 0 | 12. 2 | 22. 0 |
| 3. 1 | 13. 0 | 23. 1 |
| 4. 1 | 14. 2 | 24. 1 |
| 5. 6 | 15. 1 | 25. 1 |
| 6. 1 | 16. 1 | 26. 2 |
| 7. 0 | 17. 1 | 27. 1 |
| 8. 4 | 18. 1 | 28. 0 |
| 9. 3 | 19. 2 | 29. 1 |
| 10. 0 | 20. 0 | |

Medium (drill)

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|-------|--------|-----------|
| 1. 14 | 9. 3 | 17. true |
| 2. 6 | 10. 4 | 18. true |
| 3. 12 | 11. 19 | 19. false |
| 4. 20 | 12. 15 | 20. true |
| 5. 16 | 13. 5 | 21. false |
| 6. 17 | 14. 10 | 22. true |
| 7. 8 | 15. 7 | 23. true |
| 8. 13 | 16. 9 | 24. false |

Challenging

- 10
- e.g. A, B, C, D, E, M, T, U, V, W, Y
- e.g. a rectangle

Extension

- a shape with 2 axes
- 12

Exit ticket

- Level 1: 3
- Level 2: 1
- Level 3: 6

6.04 Rotational symmetry (booklet p. 21)

WE DO examples (teacher)

- Ex 1a: 2
- Ex 1b: 4
- Ex 2a: 2
- Ex 2b: none (order 1)
- Ex 3a: 2

6. Ex 3b: 72°

Easy (drill)

1. 2	11. 2	21. 1
2. 4	12. 2	22. 2
3. infinite	13. 1	23. 1
4. 2	14. 1	24. 2
5. 1	15. 2	25. 1
6. 5	16. 1	26. 1
7. 1	17. 1	27. 2
8. 1	18. 1	28. 1
9. 2	19. 1	29. 1
10. 1	20. 2	

Medium (drill)

1. order 9, 40°	9. order 24, 15°	17. true
2. order 3, 120°	10. order 20, 18°	18. true
3. order 36, 10°	11. order 15, 24°	19. false
4. order 10, 36°	12. order 45, 8°	20. true
5. order 40, 9°	13. order 5, 72°	21. true
6. order 18, 20°	14. order 4, 90°	22. true
7. order 6, 60°	15. order 12, 30°	23. false
8. order 8, 45°	16. true	

Challenging

1. e.g. H, I, N, O, S, X, Z
2. Yes, e.g. a parallelogram or the letter S
3. 5

Extension

1. 12
2. order 6: 60° is larger than 45°

Exit ticket

1. Level 1: 4
2. Level 2: no
3. Level 3: 60°

6.05 Classifying triangles (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: equilateral, acute-angled
2. Ex 1b: scalene, right-angled
3. Ex 2a: isosceles, acute-angled; $\angle F = \angle H$
4. Ex 2b: scalene, obtuse-angled
5. Ex 3a: scalene, acute-angled
6. Ex 3b: equilateral, acute-angled

Easy (drill)

1. obtuse	11. acute	21. isosceles
2. acute	12. acute	22. equilateral
3. obtuse	13. scalene	23. equilateral
4. acute	14. equilateral	24. isosceles
5. acute	15. isosceles	25. isosceles
6. obtuse	16. scalene	26. isosceles
7. acute	17. isosceles	27. scalene
8. acute	18. scalene	28. scalene
9. acute	19. equilateral	
10. obtuse	20. isosceles	

Medium (drill)

1. isosceles, acute	8. isosceles, acute	16. yes
2. equilateral, acute	9. isosceles, acute	17. yes
3. isosceles, acute	10. scalene, obtuse	18. no
4. isosceles, right	11. scalene, obtuse	19. no
5. scalene, obtuse	12. scalene, obtuse	20. no
6. isosceles, acute	13. scalene, obtuse	21. yes
7. scalene, obtuse	14. scalene, obtuse	22. yes
	15. isosceles, acute	23. yes

Challenging

1. scalene, acute-angled
2. Yes: isosceles right-angled (the 7 cm side is the longest)
3. isosceles, obtuse-angled

Extension

1. 40° and 40°
2. equilateral

Exit ticket

1. Level 1: equilateral
2. Level 2: isosceles, right-angled
3. Level 3: isosceles, obtuse-angled

6.06 Angle sum of a triangle (booklet p. 33)

WE DO examples (teacher)

1. Ex 1a: $x = 59$
2. Ex 1b: $u = 51$
3. Ex 2a: $a = 127$
4. Ex 2b: $y = 50$, $z = 80$
5. Ex 3a: $n = 62$, $m = 56$
6. Ex 3b: $\angle PQR = 36^\circ$ (C)

Easy (drill)

1. 16°	11. 39°	21. 29°
2. 75°	12. 53°	22. 33°
3. 84°	13. 79°	23. 15°
4. 62°	14. 72°	24. 43°
5. 129°	15. 12°	25. 62°
6. 25°	16. 16°	26. 23°
7. 133°	17. 52°	27. 67°
8. 37°	18. 51°	28. 42°
9. 42°	19. 80°	
10. 79°	20. 26°	

Medium (drill)

1. 78° and 78°	9. apex 102° , other base 39°	16. 49° and 49°
2. apex 30° , other base 75°	10. apex 72° , other base 54°	17. $x = 21$
3. apex 70° , other base 55°	11. apex 138° , other base 21°	18. $x = 48$
4. 65° and 65°	12. apex 36° , other base 72°	19. $x = 35$
5. apex 130° , other base 25°	13. apex 104° , other base 38°	20. $x = 16$
6. 15° and 15°	14. apex 22° , other base 79°	21. $x = 47$
7. apex 140° , other base 20°	15. apex 88° , other base 46°	22. $x = 47$
8. apex 118° , other base 31°		23. $x = 27$
		24. $x = 15$
		25. $x = 25$
		26. $x = 64$
		27. $x = 33$
		28. $x = 19$

Challenging

1. $x = 30$
2. 72° each
3. No: $100 + 90$ is already more than 180

Extension

1. they form a straight line (180°)
2. two obtuse angles already add to more than 180°

Exit ticket

1. Level 1: 70°
2. Level 2: 62°
3. Level 3: 46°

6.07 Exterior angle of a triangle (booklet p. 39)

WE DO examples (teacher)

1. Ex 1a: $x = 115$
2. Ex 1b: $x = 111$
3. Ex 2a: $x = 112$
4. Ex 2b: $p = 96$
5. Ex 3a: $m = 99$, $n = 81$
6. Ex 3b: it is the sum of both, so it is bigger than either

Easy (drill)

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|-----------------|-----------------|-----------------|
| 1. 94° | 11. 104° | 21. 52° |
| 2. 124° | 12. 134° | 22. 114° |
| 3. 87° | 13. 37° | 23. 125° |
| 4. 139° | 14. 34° | 24. 56° |
| 5. 94° | 15. 78° | 25. 160° |
| 6. 72° | 16. 105° | 26. 139° |
| 7. 88° | 17. 62° | 27. 158° |
| 8. 89° | 18. 102° | 28. 136° |
| 9. 80° | 19. 131° | |
| 10. 139° | 20. 87° | |

Medium (drill)

- | | | |
|----------------|-----------------|--------------|
| 1. 81° | 11. 143° | 21. $x = 58$ |
| 2. 93° | 12. 84° | 22. $x = 70$ |
| 3. 92° | 13. 76° | 23. $x = 66$ |
| 4. 33° | 14. 130° | 24. $x = 25$ |
| 5. 25° | 15. 82° | 25. $x = 37$ |
| 6. 17° | 16. 25° | 26. $x = 20$ |
| 7. 56° | 17. $x = 17$ | 27. $x = 43$ |
| 8. 54° | 18. $x = 79$ | 28. $x = 75$ |
| 9. 11° | 19. $x = 57$ | |
| 10. 13° | 20. $x = 36$ | |

Challenging

1. $x = 37$
2. 70° , 70° , 40°
3. 360°

Extension

1. 144°
2. 55° and 75°

Exit ticket

1. Level 1: 115°
2. Level 2: 65°
3. Level 3: 40°

6.08 Quadrilaterals (booklet p. 45)

WE DO examples (teacher)

1. Ex 1a: non-convex
2. Ex 1b: parallelogram
3. Ex 2a: rhombus
4. Ex 2b: trapezium
5. Ex 3a: a parallelogram with all sides equal
6. Ex 3b: trapezium

Easy (drill)

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|------------------|-----------|-----------|
| 1. kite | 6. square | 12. false |
| 2. rhombus | 7. true | 13. false |
| 3. rectangle | 8. true | 14. false |
| 4. isosceles | 9. true | 15. true |
| trapezium | 10. true | 16. false |
| 5. parallelogram | 11. false | |

Medium (drill)

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|---------------|--|-------------------|
| 1. 1 pair, 0 | 5. 1 pair, 0 (a right trapezium has 2) | 8. kite |
| 2. 2 pairs, 4 | | 9. rectangle |
| 3. 2 pairs, 4 | | 10. parallelogram |
| 4. 2 pairs, 0 | 6. 0 pairs, 0 | 11. square |
| | 7. 2 pairs, 0 | 12. rhombus |

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|-------------|-------------------------|------------|
| 13. rhombus | 14. isosceles trapezium | 15. square |
|-------------|-------------------------|------------|

Challenging

1. parallelogram
2. a parallelogram with a right angle
3. Yes: it is a parallelogram with all sides equal

Extension

1. a kite is separate; yes, a rhombus is a kite
2. rectangle and rhombus

Exit ticket

1. Level 1: yes
2. Level 2: trapezium
3. Level 3: a quadrilateral with both pairs of opposite sides parallel

6.09 Angle sum of a quadrilateral (booklet p. 51)

WE DO examples (teacher)

1. Ex 1a: $y = 70$
2. Ex 1b: $x = 81$
3. Ex 2a: $m = 26$
4. Ex 2b: $y = 59$
5. Ex 3a: a diagonal splits it into 2 triangles: $2 \times 180^\circ$
6. Ex 3b: 90° each

Easy (drill)

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|----------------|--|--|
| 1. 54° | 8. 56° | 15. 157° , 23° , 157° |
| 2. 40° | 9. 144° | 16. 149° , 31° , 149° |
| 3. 76° | 10. 69° , 111° , 69° | 17. 34° , 146° , 34° |
| 4. 59° | 11. 154° , 26° , 154° | 18. 153° , 27° , 153° |
| 5. 106° | 12. 68° , 112° , 68° | 19. 144° , 36° , 144° |
| 6. 63° | 13. 138° , 42° , 138° | 20. 74° , 106° , 74° |
| 7. 80° | 14. 61° , 119° , 61° | 21. 20° , 160° , 20° |

Medium (drill)

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|--------------|---------------|--------------|
| 1. $x = 97$ | 9. $x = 100$ | 17. $x = 51$ |
| 2. $x = 128$ | 10. $x = 129$ | 18. $x = 23$ |
| 3. $x = 103$ | 11. $x = 114$ | 19. $x = 15$ |
| 4. $x = 109$ | 12. $x = 96$ | 20. $x = 27$ |
| 5. $x = 85$ | 13. $x = 50$ | 21. $x = 29$ |
| 6. $x = 77$ | 14. $x = 24$ | 22. $x = 40$ |
| 7. $x = 66$ | 15. $x = 53$ | 23. $x = 36$ |
| 8. $x = 76$ | 16. $x = 10$ | 24. $x = 79$ |

Challenging

1. 36° , 72° , 108° , 144°
2. No: $4 \times$ more than 90° is more than 360°
3. 122° , 58° , 122°

Extension

1. 720°
2. 108°

Exit ticket

1. Level 1: 110°
2. Level 2: 80°
3. Level 3: 115° , 65° , 115°

6.10 Properties of quadrilaterals (booklet p. 57)

WE DO examples (teacher)

1. Ex 1a: e.g. opposite sides equal; opposite angles equal; diagonals bisect each other
2. Ex 1b: C

3. Ex 2a: rectangle, square
4. Ex 2b: rectangle, square
5. Ex 3a: parallelogram, rectangle, rhombus, square
6. Ex 3b: parallelogram, rectangle, rhombus, square

Easy (drill)

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|---------------|---------------|---------------|
| 1. true | 8. square, | 10. square, |
| 2. true | rectangle, | rectangle, |
| 3. true | rhombus, | isosceles |
| 4. false | parallelogram | trapezium |
| 5. false | 9. square, | 11. square, |
| 6. true | rhombus | rectangle, |
| 7. square, | | rhombus, |
| rhombus, kite | | parallelogram |
| | | 12. trapezium |

Medium (drill)

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|----------------|----------------|----------------|
| 1. 50° | 5. 13 cm | 9. 125° |
| 2. 28 cm | 6. 110° | 10. 28 cm |
| 3. 100° | 7. 12 cm | 11. 12 cm |
| 4. 65° | 8. 90° | 12. 9 cm |

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|---|---|--|
| 13. it has 4 equal sides | 15. both pairs of opposite sides are parallel | 17. it has 4 right angles and opposite sides equal |
| 14. both pairs of opposite sides are parallel | 16. it has two pairs of equal adjacent sides | 18. it has two pairs of equal adjacent sides |

Challenging

1. $CD = 7$ cm, $AD = 4$ cm, $\angle C = 65^\circ$
2. 90°
3. square

Extension

1. 5 cm
2. the diagonal bisects each 90° angle

Exit ticket

1. Level 1: rectangle or square
2. Level 2: they bisect each other at right angles
3. Level 3: rectangle (or square)